

Explicit map $\Phi_2(s)$

	AA	AB	BA	BB
ABAB	0	2	1	0
AABB	1	1	0	1
BABA	0	1	2	0

length-2 substring coordinate

Kernel $K_2 = \Phi_2 \Phi_2^\top$

	ABAB	AABB	BABA
ABAB	5	2	4
AABB	2	3	1
BABA	4	1	5

ABAB and BABA are close because both activate AB and BA; no string enumeration beyond observed windows is needed.